Supplementary Material

Reframing preprocessing selection as model-internal calibration in near-infrared spectroscopy

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Contents

1 Purpose of the Supplement

This supplement documents the evidence chain behind the main manuscript. It is deliberately more operational than the article: local result sources are listed, claims are mapped to files, deployable and oracle quantities are separated, and reviewer-sensitive weaknesses are made explicit. The goal is to make the AOM study auditable rather than to hide ongoing work behind a single compressed results table.

The supplement uses local artifacts generated in the repository, including the refreshed cohort manifest, partial aggregation outputs and the seeds 0/1/2 AOM-PLS robustness run.

For a Talanta submission, the supplement should ultimately become the place where full dataset tables, paired statistical tests, operator-selection frequencies, failure cases, leakage audits and software reproducibility metadata are made explicit. The present version already establishes that structure and marks the remaining experiments required before submission.

2 Notation and Linear-Operator Scope

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ be a centered spectral matrix and $\mathbf{Y} \in \mathbb{R}^{n\times q}$ a centered response matrix. A strict linear spectral operator $\mathbf{A}_b \in \mathbb{R}^{p\times p}$ acts as

$$\mathbf{X}_b = \mathbf{X}\mathbf{A}_b^\top. \tag{1}$$

The operator bank always contains the identity. This makes standard PLS or standard Ridge available to the selection procedure.

The strict-linear scope includes smoothing filters, derivative filters, finite differences and detrending projections. It does not include sample-adaptive or fitted transformations such as SNV, MSC, EMSC or ASLS. Those transformations are useful in NIRS but must be treated as fold-local pipeline branches. This distinction is central to the paper because the fast AOM algebra is valid only for fixed operators inside a calibration fold.

3 AOM-PLS Derivation

3.1 Covariance identity

For a fixed linear operator $\mathbf{A}_b$, the transformed cross-covariance is

$$\mathbf{X}_b^\top\mathbf{Y} = (\mathbf{X}\mathbf{A}_b^\top)^\top\mathbf{Y} = \mathbf{A}_b\mathbf{X}^\top\mathbf{Y}. \tag{2}$$

This identity permits candidate operators to be evaluated in covariance space for SIMPLS and in adjoint form for NIPALS. It avoids materializing every transformed matrix $\mathbf{X}\mathbf{A}_b^\top$ when the operator can be applied directly to vectors or matrices.

3.2 Original-space coefficients

If a PLS component is estimated in transformed space with direction r_a, the corresponding original-space direction is

$$z_a = \mathbf{A}_b^\top r_a. \tag{3}$$

Collecting k such directions gives $\mathbf{Z} = [z_1, \ldots, z_k]$ and scores $\mathbf{T} = \mathbf{X}\mathbf{Z}$. The final coefficient matrix is

$$\mathbf{B} = \mathbf{Z}(\mathbf{P}^\top \mathbf{Z})^{+}\mathbf{Q}^\top. \tag{4}$$

Thus AOM-PLS remains a single linear calibration model on the original wavelength grid.

4 AOM-Ridge Derivation

For centered $\mathbf{X}_c$ and $\mathbf{Y}_c$, Ridge regression can be solved in the dual. A single operator gives

$$\mathbf{Z}_b = \mathbf{X}_c \mathbf{A}_b^\top, \tag{5}$$

$$\mathbf{K}_b = \mathbf{Z}_b \mathbf{Z}_b^\top = \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top, \tag{6}$$

$$\mathbf{C} = (\mathbf{K}_b + \alpha \mathbf{I}_n)^{-1}\mathbf{Y}_c. \tag{7}$$

The original-space coefficient matrix is

$$\boldsymbol{\beta}_b = \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top \mathbf{C}. \tag{8}$$

For a superblock or weighted operator mixture,

$$\mathbf{K} = \sum_{b=1}^{m} s_b^2 \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top, \qquad \mathbf{M} = \sum_{b=1}^{m} s_b^2 \mathbf{A}_b^\top \mathbf{A}_b, \tag{9}$$

and the original-space coefficient matrix is

$$\boldsymbol{\beta} = \mathbf{M}\mathbf{X}_c^\top (\mathbf{K} + \alpha \mathbf{I}_n)^{-1}\mathbf{Y}_c. \tag{10}$$

5 Algorithmic Sketches

5.1 Global AOM-PLS

1. Build a compact operator bank containing identity and a small set of smoothing, derivative and detrending operators.
2. For each candidate number of components and each operator, evaluate calibration-fold performance.
3. Select the operator and component count with the best calibration-fold criterion.
4. Refit on the complete calibration set.
5. Store selected operator, component count, original-space coefficients, centering constants and fold-level diagnostics.

5.2 AOM-Ridge

1. Build operator-induced kernels $\mathbf{K}_b = \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top$ inside the calibration fold.
2. Select α and operator geometry by nested calibration-fold validation.
3. Solve the dual Ridge system.
4. Recover original-space coefficients.
5. Refit on the complete calibration set and evaluate once on the external test set.

Table 1: Strict operators and branch preprocessors considered in the AOM study.

Group	Examples	Treatment in the paper
Identity	I	Always included; recovers the standard model.
Smoothing	Savitzky–Golay smoothing	Strict linear operator.
Derivatives	SG first and second derivatives, finite difference	Strict linear operator.
Detrending	degree-1 and degree-2 trend projections	Strict linear operator when fitted inside folds.
Scatter correction	SNV, MSC, EMSC	Fold-local branch preprocessing, not a fixed operator.
Baseline correction	ASLS and related smoothers	Fold-local branch preprocessing, not a fixed operator.
Orthogonal correction	OSC-style branches	Fold-local branch; not part of the strict AOM algebra.

6 Operator and Branch Catalogue

7 Empirical Path to the Compact Operator Bank

The compact operator bank used in the main manuscript is not an arbitrary small list of preprocessors. It is the result of a sequence of AOM-specific experiments whose common conclusion is that, for NIRS calibration with small external-validation splits, the main risk is not insufficient preprocessing variety but selection variance over too many near-duplicate operators. The compact bank keeps one or two representatives of the main linear spectral families while avoiding large redundant grids of Savitzky–Golay widths, Norris–Williams variants and composed operators.

7.1 Bank genealogy

Table 2: Operator-bank genealogy used during AOM development. Candidate pairs are counted as bank size $\times K_{\max}$ with $K_{\max} = 15$, before fold replication.

Bank	Ops	Candidate pairs	Purpose in the study
`compact`	9	135	Minimal strict-linear bank used for the final deployable AOM-PLS recommendation and for the C++ `AOM_lib` implementation.
`family_pruned`	15	225	Pruned version of the production bank, keeping at most two representatives per preprocessing family. It tests whether a small but broader family set beats the compact bank.
`response_dedup`	46–47	690–705	Default-bank reduction by intrinsic-response cosine similarity. It tests whether removing near-duplicate operator responses is enough to control selection variance.

Bank	Ops	Candidate pairs	Purpose in the study
`default`	100	1500	Mirror of the deployed `nirs4all` production bank: identity, 32 Savitzky–Golay/Norris–Williams/finite-difference operators, three detrend projections and 64 SG-by-detrend compositions.
`deep3`	116	1740	Default bank plus third-order strict-linear composition chains. Used to test whether deeper composed preprocessors add useful capacity.
`deep4`	132	1980	Default bank plus third- and fourth-order composition chains. Used as a stress test for bank-size effects.
`compact+FCK`	17	255	Compact bank augmented with eight fractional convolution kernels. Used as a diagnostic extension, not promoted as the default.

The final compact bank is exactly:

$$\begin{aligned}\mathcal{A}_{\text{compact}} = \{&\text{identity}, \text{SG}^{(0)}_{11,2}, \text{SG}^{(0)}_{21,3}, \text{SG}^{(1)}_{11,2}, \text{SG}^{(1)}_{21,3},\\ &\text{SG}^{(2)}_{11,2}, \text{detrend}_1, \text{detrend}_2, \text{FD}_1\}.\end{aligned} \tag{11}$$

The C++ implementation uses the same ordering: `identity`, `sg_smooth_w11_p2`, `sg_smooth_w21_p3`, `sg_d1_w11_p2`, `sg_d1_w21_p3`, `sg_d2_w11_p2`, `detrend_d1`, `detrend_d2`, `fd_d1`.

7.2 Why fewer operators can be better

For a bank with B operators and $K_{\max}$ component prefixes, a global AOM selector evaluates approximately $M = BK_{\max}$ operator/component candidates per selection split, and FM candidates under F-fold cross-validation. On small calibration sets, validation RMSE estimates have non-negligible noise. The empirical minimum over many candidates is therefore optimistically biased; a crude order-statistic approximation is

$$\mathbb{E}[\min_i \widehat{R}_i] \approx \mu - \frac{\sigma}{\sqrt{n_{\text{val}}}}\sqrt{2\log M}. \tag{12}$$

Thus adding near-duplicate operators can worsen external performance even when the larger bank strictly contains the smaller one. In the AOM setting, moving from `compact` to `default` increases M from 135 to 1500 at $K_{\max} = 15$. The nominal selection-bias factor $\sqrt{\log(1500)/\log(135)}$ is about 1.22, which is large enough to matter for BEER- or BISCUIT-sized splits where the internal validation set contains only a few samples. The compact bank was therefore treated as a bias-control device, not only as a speed optimization.

7.3 Ablation evidence

Table 3: AOM-PLS ablations that led to the compact-bank recommendation. Lower median relative RMSEP is better. Values are from `bench/AOM_v0/publication/tables/relative_rmsep_per_variant.csv`; full variant identifiers are preserved in that source table.

Variant	Bank / selector	N	Median	Wins	Median fit (s)
nirs4all default	100-op production holdout	57	0.999	29	0.96
Default NIPALS	100-op default holdout	57	0.999	29	8.44
Family-pruned	15-op holdout	57	0.998	30	0.58
Response-dedup	46–47-op holdout	57	0.998	29	1.97
Compact CV-3	9-op CV-3	57	0.997	32	0.96
Compact CV-5	9-op CV-5	57	0.992	38	1.23
Compact repeated CV-3	9-op repeated CV-3	57	0.984	39	2.08
Response-dedup CV-3	46–47-op CV-3	57	0.988	33	4.27
ASLS compact CV-3	ASLS + 9-op CV-3	57	0.979	38	0.91
ASLS compact repeated CV-3	ASLS + 9-op repeated CV-3	57	0.975	39	2.21
ASLS compact CV-5	ASLS + 9-op CV-5	57	0.960	42	1.36
ASLS family-pruned CV-3	ASLS + 15-op CV-3	57	0.964	38	1.60
ASLS response-dedup CV-3	ASLS + 46–47-op CV-3	57	0.964	37	4.57
SPXY response-dedup CV-3	SPXY + 46–47-op CV-3	54	0.962	35	29.16

These ablations support three practical conclusions. First, the compact bank already matches the 100-operator production/default bank at the median while using roughly one tenth of the operator/component candidates. Second, the largest gain comes from replacing deterministic holdout selection with fold-based selection: compact CV-5 improves from the production/default median near 0.999 to 0.992 without any ASLS branch, and repeated CV-3 reaches 0.984. Third, adding the ASLS branch before compact AOM-PLS gives the current headline variant, but the nearby 15-op and 46–47-op bank results show that the conclusion is not a one-off artifact of one tiny bank. The compact bank is selected as the deployable default because it lies on the accuracy/time frontier: ASLS + response-dedup with SPXY can be marginally close, but it is substantially slower and has fewer paired rows.

7.4 Selected operators inside the compact bank

The selected-operator distribution is important for interpretation. The identity remains selected on several datasets, which is expected when raw spectra already carry the analyte signal. The most frequent non-identity operator is the wider SG smoother, followed by derivative and detrending operators. This supports the design choice that the compact bank should cover distinct spectral mechanisms rather than dense parameter grids within one mechanism.

7.5 Extensions that were not promoted

Several extensions were explored and intentionally not moved into the compact default.

- The 100-operator default bank and the deep3/deep4 banks did not improve the median

Table 4: Selected first operator in the compact bank on the 57-dataset cohort. Counts are taken from the fold-based AOM-PLS result files and show that the compact bank does not collapse to a single preprocessing choice.

Operator	Spectral role	AOM CV-5	ASLS-AOM CV-5
`identity`	Raw centered spectrum / PLS fall-back	5	7
`sg_smooth_w11_p2`	Short-window SG smoothing	0	1
`sg_smooth_w21_p3`	Wider SG smoothing	12	17
`sg_d1_w11_p2`	Short-window first derivative	9	7
`sg_d1_w21_p3`	Wider first derivative	8	7
`sg_d2_w11_p2`	Second derivative	1	3
`detrend_d1`	Linear detrending projection	8	3
`detrend_d2`	Quadratic detrending projection	8	4
`fd_d1`	Finite difference first derivative	6	8

external RMSEP enough to justify the added selection variance and runtime. Deep banks mostly reproduced the default-bank choice on the tested splits.

- OSC and EMSC branches were not promoted as defaults. Existing AOM_v0 summaries show OSC variants hurting at the default component setting and EMSC degree 1–2 being numerically unstable on small splits.

- SNV can help as a branch, but SNV+ASLS did not improve over ASLS alone in the 57-dataset summary. SNV, MSC, EMSC and ASLS remain branch preprocessors, not strict fixed operators in the AOM algebra.

- Fractional convolution kernel operators were implemented as diagnostic bank extensions (`compact+FCK`: 9 compact operators plus eight FCK kernels). They remain useful evidence that fractional-derivative preprocessors were considered, but they are not part of the current compact recommendation until they pass the same paired, multi-seed and no-harm gates as the compact AOM-PLS variants.

8 Benchmark Protocol Ledger

Table 5: Local evidence map used to prepare the manuscript.

Artifact	Content	Use in the paper
`bench/AOM_v0/Summary.md`	57-dataset AOM-PLS summary; compact ASLS+AOM CV-5 champion.	Primary AOM-PLS headline source.
`bench/AOM_v0/publication/tables/relative_rmsep_per_variant.csv`	Relative RMSEP and fit-time summaries for AOM-PLS variants.	Source for median RMSEP/PLS, wins and fit time.
`bench/AOM_v0/docs/AOMPLS_MATH_SPEC.md`	AOM-PLS operator algebra, NIPALS/SIMPLS conventions.	Method verification and notation.
`bench/AOM_v0/docs/AOMPLS_VALIDATION.md`	Equivalence tests: identity, fixed operator, NumPy/Torch parity.	Implementation validation.

Artifact	Content	Use in the paper
bench/AOM_v0/Ridge/docs/AOM_RIDGE_MATH_SPEC.md	Dual AOM-Ridge derivation and branch warnings.	Ridge method section.
bench/AOM_v0/Ridge/publication/tables/table_per_method_summary.tex	Deployable AOM-Ridge variants on 52 datasets.	Conservative deployable Ridge claim.
bench/AOM_v0/Ridge/publication/tables/table_summary.tex	Oracle envelope against reference baselines.	Upper-bound Ridge claim only.
bench/AOM_v0/Ridge/docs/HEADLINE_SPXY3_NESTED_AUDIT.md	Nested-selection audit for headline Ridge variants.	Leakage-risk discussion.
bench/tabpfn_paper/article/main.tex	Reference benchmark protocol, preprocessing search budgets and splits.	Baseline protocol context; model name not emphasized in the article.
bench/benchmark_master_results.csv	Unified registry of many variants and maturity levels.	Exploration registry, not a direct source of all headline statistics.

8.1 Cohort manifest

The frozen review manifest is stored at `paper_aom/review/cohort_manifest.csv`. It contains 78 source rows: 61 regression rows and 17 classification rows. For primary paired analyses, the effective cohort is 60 regression datasets plus 16 classification datasets: the QUARTZ regression row is retained for absolute-error and failure tables but excluded from ratio summaries because the reference denominator is near zero, and one classification row is excluded because source files are missing. The companion denominator document `paper_aom/review/cohort_manifest.md` records the gates for TabPFN-allowed subsets and AOM-Ridge-global allowability.

8.2 Software artefacts

Table 6 mirrors the software-status table used in the main manuscript. It separates production `nirs4all` code, the new AOM-lib-backed wrapper, parity-verified C++/Python `AOM_lib` components, code-ready bindings that were not locally executed in this validation environment, and AOM-Ridge research code.

8.3 Multi-seed regression robustness

The refreshed AOM-PLS robustness run is `bench/scenarios/runs/paper_aom_aompls_seeds012/results.csv`. It covers 55 datasets, 9 variants and seeds 0/1/2, with 1486 CSV lines at validation time. For the headline comparison `ASLS-AOM-compact-cv5-numpy` versus `PLS-standard-numpy`, the 53 paired datasets give a median RMSEP/PLS ratio of 0.9623 and 37/53 wins after averaging RMSEP over seeds. This is close to the earlier single-seed median 0.960, but the win denominator changes from 57 to 53 under the refreshed paired multi-seed run.

8.4 Parity validation

The dedicated `AOM_lib` implementation was validated against the reference Python AOM_v0 calculations on BEER, CORN and ALPINE under three split/selection regimes: kfold5, kfold5+oneSE and spxy5. The local validation commands `cpp/build/test_operators`, `cpp/build/test_parity_kfoldcpp/tests/reference` and `python/tests/test_parity.py` passed. Across the

Table 6: Software artefacts distributed with the AOM study.

Component	Language	Lines	Tests	Status
`nirs4all` AOM-PLS (reference)	Python	1,600	`tests/unit/operators/models/test_aom_pls*`	*production*
`nirs4all` AOM-lib wrapper	Python	50	`tests/unit/operators/models/test_aom_pls_a`	*integration (new)*
`AOM_lib` C++17 core	C++17	1,400	`cpp/build/test_operators`, `test_parity_kfold`	*parity-verified*
`AOM_lib` C ABI	C/C++ `extern "C"`	250	`cpp/build/test_smoke`	*smoke-tested*
`AOM_lib` Python binding	Python + pybind11	400	`python/tests/test_parity.py`	*parity-verified* (9/9)
`AOM_lib` R binding	R + Rcpp	600	`r/aompls/tests/testthat/test-parity.R`	*code-ready, not tested locally*
`AOM_lib` MATLAB binding	MATLAB + MEX	400	`matlab/test_parity.m`	*code-ready, not tested locally*
`AOM_lib` Julia binding	Julia + `ccall`	300	`julia/AomPLS/test/runtests.jl`	*code-ready, not tested locally*
`AOM_lib` JS/WASM binding	JavaScript + Emscripten	400	`js/test/parity.mjs`	*code-ready, not tested locally*
AOM-Ridge research code	Python	8,800	`bench/AOM_v0/Ridge/tests/`	*research*

Status legend: *production* – shipping in `nirs4all` releases; *integration* – new wrapper exposing the C++ core through the Python pipeline API; *parity-verified* – numerical agreement with the Python AOM_v0 reference on BEER/CORN/ALPINE × kfold5/kfold5+oneSE/spxy5 (coefficient tolerance $< 10^{-8}$, prediction tolerance $< 10^{-8}$, RMSE-curve tolerance $< 10^{-9}$); *smoke-tested* – builds and runs a minimal end-to-end fit/predict against the C ABI; *code-ready, not tested locally* – bindings compile and ship with parity tests, but the host interpreter/toolchain (R, MATLAB, Julia, Emscripten) was not available in the current validation environment; *research* – experimental code distributed under `bench/AOM_v0` pending API stabilization. All artifacts live in the public repository under `bench/AOM_lib` and `bench/AOM_v0`.

Table 7: Seed-stability summary emitted by `paper_aom/review/final_stats.md`. Median and IQR are the raw RMSEP dispersion summaries across available seeds, so they are used here as stability diagnostics rather than cross-trait effect sizes.

Variant	Datasets	Seeds	Median	IQR
ASLS-AOM-compact-cv5-numpy	59	3	0.4515	3.3055
AOM-compact-cv5-numpy	59	3	0.6327	3.3118
AOMRidge-Blender-headline-spxy3	20	3	0.3300	2.2792
AOMRidge-Local-compact-knn50	20	3	0.5549	2.5818
AOMRidge-AutoSelect-headline-spxy3	20	3	0.3455	2.2845
AOMRidge-global-compact-none	20	3	0.4153	2.7123
Ridge-tabpfn-hpo-60trials	7	3	0.4257	3.2861
PLS-default-cv5	7	3	0.5713	3.4682
PLS-tabpfn-hpo-25trials	7	3	0.3535	4.1037

nine parity cases, the largest reported coefficient difference was below 10^{-8}, with prediction and RMSE-curve differences also below the tolerance used by the parity tests.

9 Claim Ledger

Table 8: Main claims and their current evidence status.

Claim	Status	Dataset count	Source / caveat
ASLS + compact AOM-PLS CV-5 improves over PLS with median RMSEP/PLS 0.9623.	Multi-seed AOM-PLS robustness result.	53	`paper_aom_aompls_seeds012/results.csv`; paired after averaging over seeds.

Claim	Status	Dataset count	Source / caveat
AOM-PLS compact CV-5 wins against PLS on 37 datasets.	Multi-seed AOM-PLS robustness result.	53	Same source; replaces the earlier single-seed 57-dataset headline.
AOM-PLS compact CV-5 median fit time is approximately 1.65 s.	Measured in refreshed aggregation.	339	`paper_aom/review/final_stats.md`; hardware- and partial-run dependent.
AOM-Ridge Blender has median RMSEP/Ridge-HPO ratio 0.905 in the refreshed overlap.	Partial regenerated evidence.	4	`paper_aom/review/final_stats.md`; denominator limited by incomplete HPO overlap.
AOM-Ridge Blender wins 4/4 against Ridge-HPO in the refreshed overlap.	Partial regenerated evidence.	4	Full-cohort Blender/AutoSelect multi-seed promotion is pending.
AOM-Ridge oracle improves over tuned Ridge by median 4.73%.	Oracle envelope only.	52	`table_summary.tex`; not a deployable method claim.
AOM-Ridge oracle wins 45/52 against tuned Ridge.	Oracle envelope only.	52	Use only as headroom / upper-bound evidence.
Large preprocessing searches evaluate thousands of fold-level candidates.	Reference protocol fact.	per dataset	PLS/Ridge approximately 5400, CNN approximately 2700, CatBoost approximately 90.

10 AOM-PLS Variant Summary

Table 9: Historical single-seed AOM-PLS variants from the existing 57-dataset summary. The refreshed multi-seed headline is reported in the robustness subsection above.

Variant	Datasets	Median RMSEP/PLS	Wins	Median fit (s)
ASLS-AOM compact CV-5	57	0.960	42	1.36
ASLS-AOM response-dedup CV-3	57	0.964	37	4.57
ASLS-AOM family-pruned CV-3	57	0.964	38	1.60
ASLS-AOM compact repeated CV-3	57	0.975	39	2.21
ASLS-AOM compact CV-3	57	0.979	38	0.91

Table 10: Single-seed AOM-Ridge deployable variants retained as background evidence before full multi-seed promotion; full local identifiers are in the source table.

Variant	Wins / N	Win-rate (%)	Mean rank	Median Δ vs Ridge (%)
Blender headline SPXY3	35/52	67.3	3.56	-2.22
AutoSelect headline SPXY3	27/52	51.9	4.61	-0.61
Global compact none	29/52	55.8	4.79	-0.63

Table 11: AOM-Ridge oracle envelope. These values describe headroom and must not be worded as deployable practitioner performance.

Baseline	Median Δ (%)	Mean Δ capped (%)	Win-rate (%)	Wins / N
Ridge	-4.73	-12.37	86.5	45/52
PLS	-7.73	-14.66	90.4	47/52
CNN	-12.56	-16.08	74.5	35/47
CatBoost	-13.27	-19.77	73.1	38/52

11 AOM-Ridge Result Separation

12 Paired Statistical Evidence

13 Leakage and Audit Controls

14 Additional Experiments Required for a Journal Submission

Table 14: Experiment backlog.

Priority	Experiment	Purpose
P0	Freeze one regression cohort manifest.	Remove denominator drift across 52/57/58/61 dataset statements.
P0	Regenerate headline tables from one manifest.	Ensure PLS, Ridge, CatBoost, CNN, AOM-PLS and AOM-Ridge are compared on a common subset.
P0	Separate deployable and oracle tables.	Prevent reviewers from interpreting oracle envelopes as methods.
P0	Add exclusion table.	Document QUARTZ-style degenerate denominators, missing CNN rows and compute skips.
P1	Full multi-seed rerun.	Quantify selection variance and win-rate uncertainty.
P1	Paired statistical tests.	Add Wilcoxon/Friedman/Nemenyi or equivalent tests on the common subset.
P1	Strict-linear vs branch ablation.	Separate gains from AOM itself and gains from ASLS/SNV/MSC branches.
P1	Operator-frequency table.	Show which spectral operations are actually selected.
P1	Runtime scaling study.	Quantify cost versus n, p, bank size and branch count.
P1	Prediction artifact export.	Enable residual plots, bias, RPD/RPIQ, and independent audit.

Priority	Experiment	Purpose
P1	Software release validation.	Maintain C++/Python parity validation and complete local execution for the R, MATLAB, Julia and JavaScript/WASM bindings before release promotion.

15 Reproducibility Checklist

1. Record repository commit hash, Python version and package versions.
2. Store the cohort manifest used for every table.
3. Store split indices for every dataset.
4. Store fold predictions and final test predictions.
5. Store selected operators, component counts, α values and branch preprocessing choices.
6. Store runtime and hardware metadata.
7. Regenerate tables from scripts rather than manually editing manuscript tables.

16 Recommended Positioning

The safest wording for the current draft is: AOM is a general linear-operator framework; AOM-PLS has the strongest deployable full-cohort evidence; AOM-Ridge is a mathematically consistent and promising extension with deployable evidence and a stronger oracle envelope, but requires full multi-seed promotion before being described as a production default. This position is scientifically stronger than claiming universal benchmark dominance.

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Table 12: Current paired statistical evidence for the two primary deployable claims. These values were computed from existing result files and should be regenerated after the final Talanta cohort manifest is frozen.

Comparison	N	Median effect	95% bootstrap CI	Wins	Wilcoxon p_{Holm}
ASLS-AOM-compact-cv5 vs PLS-TabPFN-HPO	7	RMSEP ratio 1.012	0.966–1.064	3/7	0.875
ASLS-AOM-compact-cv5 vs PLS-default	7	RMSEP ratio 0.964	0.474–0.993	6/7	0.766
AOM-compact-cv5 vs PLS-default	7	RMSEP ratio 0.963	0.536–1.006	5/7	0.875
AOMRidge-global-compact-none vs Ridge-TabPFN-HPO	4	RMSEP ratio 1.232	0.870–1.452	1/4	0.875
AOMRidge-Local-compact-knn50 vs Ridge-TabPFN-HPO	4	RMSEP ratio 1.103	0.986–2.148	1/4	0.875
AOMRidge-Blender vs Ridge-TabPFN-HPO	4	RMSEP ratio 0.905	0.777–0.964	4/4	0.766
AOMRidge-AutoSelect vs Ridge-TabPFN-HPO	4	RMSEP ratio 0.899	0.799–0.968	4/4	0.766

Table 13: Leakage controls that must be preserved in any final rerun.

Control	Requirement
External test set	Never used for operator, branch, component or α selection.
Original splits	Preserved when supplied by the dataset source.
Generated splits	Deterministic SPXY or stratified variants, with split type logged.
Branch preprocessing	Fitted inside each calibration fold only.
Operator selection	Performed using calibration-fold validation only.
Final refit	Refit selected configuration on the full calibration set, then evaluate once on test.
Oracle tables	Clearly labeled as retrospective upper bounds.
Skipped datasets	Logged with inclusion/exclusion reason.

[7] R. J. Barnes, M. S. Dhanoa, and Susan J. Lister. Standard normal variate transformation and de-trending of near-infrared diffuse reflectance spectra. *Applied Spectroscopy*, 43(5):772–777, 1989.

[8] Paul Geladi, Donald MacDougall, and Harald Martens. Linearization and scatter-correction for near-infrared reflectance spectra of meat. *Applied Spectroscopy*, 39(3):491–500, 1985. doi: 10.1366/0003702854248656.